



Fundamentals of OOP

1. Why OOP
2. Concept of class and object
3. Principles of OOP

- Benefits programming
 - Better modelling of real life scenario in software system
 - Better security of data
 - Easy maintenance
 - Easy to extend in future as per changing needed

❏ Object

- Is a real life entity having
 1. State:
Values assigned to attributes of object
 2. Identity
Unique attribute that identified object
 3. Behavior
Response to external messages
 4. Responsibility
Overall purpose of object in the system
- Can be tangible or non-tangible
- Example:
Car, Book, Person, Date, Lecture, Exam, Question paper etc.

Class

- Is a template that models state and behavior of object
- First class is created. Then objects are created from class
- Object creation is called as object instantiation
- Resources are not allocated when class is created. They are allocated to objects that are created from class

- ❑ Major principles
 - Abstraction
 - Encapsulation
 - Inheritance
 - Polymorphism
- ❑ Minor principles
 - Strong typing
 - Persistence
 - Concurrency

- ❑ Major principles
 - Abstraction:
 - Selecting necessary details. Ignore un-necessary details.
 - It is selective ignorance
 - Encapsulation:
 - Binding state with behavior
 - Hiding implementation details from user.
 - Inheritance
 - Reusing existing class to create more special type of it
 - Creates is-a relationship
 - Polymorphism
 - Different implementation of same message by various objects
 - Real life examples

- ❑ Minor principles
 - Strong typing:
Information must be given in a specific type. Only value is not enough.
 - Persistence:
Maintaining state of object permanently, so that it can be retrieved later in a different session
Can be stored in file, database or any other device
 - Concurrency
A system having multiple sub-systems running concurrently
Example: Car with transmission, music system, AC running at a time